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To the Editors,
[Journal name]

Dear Editors,

Please find attached our manuscript “Boundary and Interface Classes for Universal Chern Hamiltonians via Doubling and Jamming” submitted for consideration. The submission package consists of the main paper (`paper_main.pdf`, 33 pages), an online supplement (`paper_supplement.pdf`, 251 pages raw, of which approximately 160 pages are load-bearing mathematics; the remaining material is internal-development audit, framing, navigation, and inventory content that a math-content reader can skip on first reading and that is explicitly flagged as such), and a companion code repository linked from the abstract. A short summary of what the paper does, and a short explanation of how we are framing it, follows.

Contribution. The paper develops a finite-dimensional doubling-and-jamming construction for boundary and interface classes of Higson–Prodan universal Chern Hamiltonians on triangulated surfaces with boundary. Given a finite triangulated oriented surface with boundary, we double it along the boundary, suppress the reflected copy by a large on-site potential, and study the resulting jammed Hamiltonian by Schur-complement analysis. The principal ingredients are: (i) the intrinsic Poincaré–Lefschetz boundary Hamiltonian and its comparison with the original-side compression of the doubled closed Hamiltonian; (ii) seven Schur-complement estimates for finite- M jamming (compressed resolvent, first-order effective resolvent, copy leakage, eigenvalue localization, contour margin, window separation, and the original-copy coupling support); (iii) a periodic triangular Bloch certificate, with closed-bulk gap, Chern signs, and integer Toeplitz interface winding all proved internally on the straight periodic interface; (iv) internal closed-bulk gap and Chern theorems for the three flat-torus benchmark scopes (triangular, hexagonal, square), unified by a general operator-algebraic Chern convergence lemma; (v) an internal Combes–Thomas exponential decay lemma; and (vi) a complete companion numerical pipeline that records finite-matrix behavior of all the mechanisms above. These ingredients are unconditional inside the manuscript: their proofs do not invoke any external closed-bulk theorem.

Delaunay-extension partial progress. Beyond the periodic flat-torus benchmark scopes of ingredient (iv), the supplement contains a dedicated Delaunay-extension chapter that develops quantitative tools for extending the internal closed-bulk analysis from periodic flat-torus triangulations to the broader quasi-uniform Delaunay class (every triangle of bounded minimum interior angle and bounded edge-length ratio, plus the Euclidean Delaunay circumdisk property). The chapter contains eighteen rigorously proved results, organized into three explicit tiers: *fourteen unconditional results* (bounded-geometry parameter lemmas, translation and vertex-perturbation coherence lemmas, inclusion-coherence and cap-product flip lemmas, single-flip and multi-flip Schur-complement

bounds, common-refinement existence in the union sense, block-isometry and block-cap-product matrix-element identities, and a quasi-isometric coherence theorem assembled by cascade); *three conditional theorems* (a Delaunay-gap stability statement, a different-dimension coherence theorem, and a three-step composition theorem) each conditional on a single shared sub-conjecture (sub-complex refinement existence in the strict inclusion sense); and *one proved negative result* (an obstruction theorem ruling out the natural “refinement-in-the-union-sense” approach to that sub-conjecture via an explicit renormalization-mismatch computation, illustrated in the cleanest possible flat-torus setting). This is partial progress, not a closure: the Delaunay-extension cluster does *not* eliminate the bounded-geometry case as an external dependency, and the conditional theorems remain conditional. What the chapter adds is a quantitatively explicit framework, with sharp constants and a documented obstruction, within which a future closure of the sub-conjecture could upgrade three conditional theorems to unconditional ones. We are presenting this chapter as honest, mathematically-careful research-medium content beyond the periodic flat-torus / hexagonal / square benchmark cases of ingredient (iv), not as a replacement for the external bounded-geometry input.

What still depends on Higson–Prodan. We are honest about the scope of the unconditional part. For the general bounded-geometry case, the headline Roe-algebra theorem identifying the doubled-and-jammed interface class $\partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X))$ still uses the Higson–Prodan closed-bulk gap and Chern theorem as an external input. The bounded-geometry closed-bulk gap and the bounded-geometry closed-bulk Chern integer are imported as a single named external hypothesis (the published Higson–Prodan PRL and arXiv articles, cited in the manuscript), and the manuscript marks every place where this hypothesis is used. The internal Combes–Thomas decay we prove gives the locality toolkit that a bounded-geometry gap proof would need, but it does not by itself supply the gap; the Delaunay-extension chapter described above gives quantitatively explicit relative-stability tools in the quasi-uniform Delaunay class but, by the documented obstruction theorem and the unresolved sub-conjecture, does not by itself close the bounded-geometry case either. Closing this last step is identified in the paper as the single research-hard external dependency and as a natural follow-up project rather than a result of the present work.

Editorial framing of the submission. The submitted manuscript is organized along this split. Part A collects the unconditional results: the finite-dimensional construction, the seven Schur-complement estimates, the periodic triangular Bloch certificate and its Toeplitz winding, the three internal flat-torus benchmark gap and Chern theorems with the unifying lemma, the internal Combes–Thomas decay, and the companion numerics. Part B states the global Roe-algebra interface theorem in its full bounded-geometry generality and the conditional integer winding for general partitions; both are conditional on the published Higson–Prodan PRL result, and we present them as a conditional framework rather than as unconditional theorems. The supplement contains an explicit Part A–Part B inventory (the iter-17 carve-out table) and a per-figure / per-table audit confirming that no numerical figure is used as a proof input and that the Part placement is consistent at the caption level. The Delaunay-extension chapter sits inside Part A as additional internal infrastructure of the unconditional kind (fourteen unconditional results) together with three conditional theorems that are flagged as conditional inside the chapter. We chose this two-part presentation deliberately: it makes the conditional dependency on the published Higson–Prodan PRL result fully visible and confines it to Part B, so the unconditional Part A material—including the Delaunay-extension unconditional content—can be evaluated on its own terms.

Why this might be of interest. The doubling-and-jamming construction itself is the main technical ingredient: it converts a closed-bulk universal Chern theorem into finite-matrix boundary and interface statements via the simplicial double, with quantitative Schur-complement control of the jamming error. The three flat-torus benchmark cases give worked closed-form examples in which the closed-bulk inputs that Higson–Prodan import from the universal model are proved internally, removing the conditional dependency entirely on those benchmark families. The periodic triangular case in particular admits an integer Toeplitz winding on the straight periodic interface, also proved internally. The Delaunay-extension chapter develops quantitative quasi-isometric relative-stability tools for the broader bounded-geometry context, with eighteen rigorously proved results across the three tiers described above and a single sharply identified remaining research obstruction. This combination—a quantitative finite-dimensional boundary framework, plus benchmark cases in which the conditional input is eliminated, plus a Delaunay-extension cluster of relative-stability tools with an honest obstruction analysis, plus an explicit cataloguing of what remains conditional—should be of interest to readers in mathematical physics and operator algebras working on bulk–interface correspondences for topological insulators.

Supplement and code. The online supplement (251 pages raw) contains the full appendix proofs for the Schur-complement estimates and the boundary comparison, the eight internal HP-replacement results (six benchmark closed-bulk gap and Chern theorems plus the operator-algebraic unification and the Combes–Thomas decay), the consolidated internal chapter, the HP-dependency inventory, the Delaunay-extension chapter with its eighteen results across the three tiers, the topological diagrams, and the numerical record (twenty-five figures and tables). The supplement also contains internal-development audit, framing, navigation, and inventory subsections that document the manuscript’s editorial history; these are marked with “skip-on-first-reading” notices at their headings, and condensed into a single ten-row master table at the start of the supplement’s reader-navigation section. A math-content reader who skips these flagged subsections encounters an effective supplement of approximately 160 pages of load-bearing mathematics; a verification-focused or editorial reader retains full access to the underlying audit trail by consulting the flagged subsections individually. The companion code repository (URL in the abstract) contains the data-generation scripts for every numerical figure and table.

Statements. The manuscript has not been published elsewhere and is not under consideration at another journal. We have no conflicts of interest to declare.

We hope the editors will find the work suitable for the journal, and we look forward to the referees’ assessment.

Sincerely,

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